Solution Requirements

Date	29 June 2026
Team ID	
Project Name	**Credit Card Approval Prediction System**
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:
Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Only authorized users can access the credit card approval prediction system using secure login credentials.	High
2	Authorization levels	The system provides role-based access for administrators and bank credit analysts to perform permitted operations.	High
3	External interfaces	Connects with the trained Machine Learning model (.pkl), Flask web application, applicant dataset, and IBM Cloud deployment services.	High
4	Transactions processing	Accepts applicant details, processes them through the ML model, and returns an approval or rejection prediction instantly.	High
5	Reporting	Displays prediction results and model performance metrics such as accuracy, confusion matrix, and classification report.	Medium
6	Business rules, etc.	Applies preprocessing, feature engineering, categorical encoding, and classification rules before generating predictions.	High
7	Compliance to laws or regulations	Protects applicant data and follows banking data privacy and security guidelines during processing.	High
8	Other	Allows future integration with banking databases and additional machine learning models.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate approval predictions with minimal processing delay.	Response time less than **2 seconds**
2	Scalability	Support increasing numbers of users and prediction requests.	Supports **1000+ prediction requests** efficiently
3	Security & Data Privacy	Secure applicant information and prevent unauthorized access.	Secure data storage and authenticated access
4	Reliability & Availability	Ensure consistent prediction service with minimal downtime.	**99% system availability**
5	Usability & Accessibility	Provide a simple, user-friendly interface for bank staff and customers.	Easy navigation with responsive web pages
6	Other	Maintain modular code for future updates and deployment.	Easy maintenance and portability